

54/74 FAMILIES OF COMPATIBLE TTL CIRCUITS

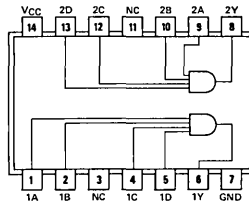
PIN ASSIGNMENTS (TOP VIEWS)

DUAL 4-INPUT POSITIVE-AND GATES

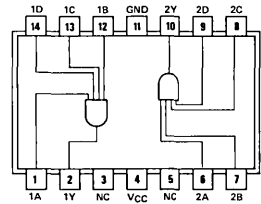
21

positive logic:
Y = ABCD

See page 6-10



SN54H21 (J) SN74H21 (J, N)
SN54LS21 (J, W) SN74LS21 (J, N)



SN54H21 (W)

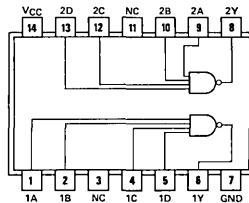
NC—No internal connection

DUAL 4-INPUT POSITIVE-NAND GATES WITH OPEN-COLLECTOR OUTPUTS

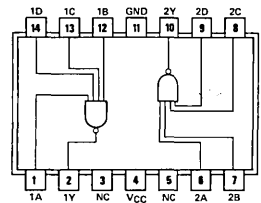
22

positive logic:
Y = ABCD

See page 6-4



SN54422 (J, W) SN74422 (J, N)
SN54H22 (J) SN74H22 (J, N)
SN54LS22 (J, W) SN74LS22 (J, N)
SN54S22 (J, W) SN74S22 (J, N)



SN54H22 (W)

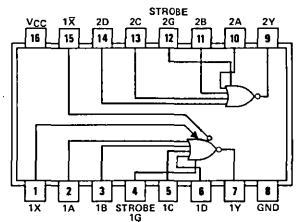
NC—No internal connection

EXPANDABLE DUAL 4-INPUT POSITIVE-NOR GATES WITH STROBE

23

positive logic:
 $1Y = \overline{1G(1A+1B+1C+1D)+X}$
 $2Y = 2G(2A+2B+2C+2D)$
X = output of SN5460/SN7460

See page 6-39



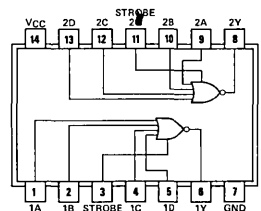
SN5423 (J, W) SN7423 (J, N)

DUAL 4-INPUT POSITIVE-NOR GATES WITH STROBE

25

positive logic:
Y = G(A+B+C+D)

See page 6-8



SN5425 (J, W) SN7425 (J, N)

5

recommended operating conditions

		54 FAMILY 74 FAMILY	SERIES 54 SERIES 74			SERIES 54H SERIES 74H			SERIES 54LS SERIES 74LS			SERIES 54S SERIES 74S			UNIT
		'08,			'H11, 'H21			'LS08, 'LS11, 'LS21			'S08, 'S11				
		MIN	NOM	MAX	MIN	NOM	MAX	MIN	NOM	MAX	MIN	NOM	MAX		
Supply Voltage, V _{CC}	54 Family	4.5	5	5.5	4.5	5	5.5	4.5	5	5.5	4.5	5	5.5	V	
	74 Family	4.75	5	5.25	4.75	5	5.25	4.75	5	5.25	4.75	5	5.25		
High-level output current, I _{OH}		-800			-500			-400			-1000			μA	
Low-level output current, I _{OL}	54 Family	16			20			4			20			mA	
	74 Family	16			20			8			20				
Operating free-air temperature, T _A	54 Family	-55		125	-55		125	-55		125	-55		125	°C	
	74 Family	0		70	0		70	0		70	0		70		

electrical characteristics over recommended operating free-air temperature range (unless otherwise noted)

PARAMETER	TEST FIGURE	TEST CONDITIONS†		SERIES 54 SERIES 74		SERIES 54H SERIES 74H		SERIES 54LS SERIES 74LS		SERIES 54S SERIES 74S		UNIT
				'08		'H11, 'H21		'LS08, 'LS11, 'LS21		'S08, 'S11		
				MIN	TYP‡ MAX	MIN	TYP‡ MAX	MIN	TYP‡ MAX	MIN	TYP‡ MAX	
V _{IH} High-level input voltage	1, 2			2		2		2		2		V
V _{IL} Low-level input voltage	1, 2			54 Family		0.8		0.8		0.8		V
				74 Family		0.8		0.8		0.8		
V _{IK} Input clamp voltage	3	V _{CC} = MIN, I _I = §		-1.5		-1.5		-1.5		-1.2		V
V _{OH} High-level output voltage	1	V _{CC} = MIN, V _{IH} = 2 V, I _{OH} = MAX		54 Family		2.4 3.4		2.4 3.4		2.5 3.4		V
				74 Family		2.4 3.4		2.7 3.4		2.7 3.4		
V _{OL} Low-level output voltage	2	V _{CC} = MIN, I _{OL} = MAX		54 Family		0.2 0.4		0.15 0.3		0.25 0.4		V
				74 Family		0.2 0.4		0.2 0.4		0.35 0.5		
		V _{IH} = 2 V I _{OL} = 4 mA		Series 74LS						0.25 0.4		
I _I Input current at maximum input voltage	4	V _{CC} = MAX		V _I = 5.5 V		1		1				mA
				V _I = 7 V				0.1				
I _{IH} High-level input current	4	V _{CC} = MAX		V _{IH} = 2.4 V		40		50				μA
				V _{IH} = 2.7 V				20		50		
I _{IL} Low-level input current	5	V _{CC} = MAX		V _{IL} = 0.4 V		-1.6		-2		-0.4		mA
				V _{IL} = 0.5 V						-2		
I _{OS} Short circuit output current♦	6	V _{CC} = MAX		54 Family		-20 -55		-40 -100		-20 -100		mA
				74 Family		-18 -55		-40 -100		-20 -100		
I _{CC} Supply current	7	V _{CC} = MAX				See table on next page						mA

See table on next page

†For conditions shown as MIN or MAX, use the appropriate values specified under recommended operating conditions.

‡All typical values are at $V_{CC} = 25^{\circ}\text{C}$.§ $I_I = -12 \text{ mA}$ for SN54/SN74*, -8 mA for SN54H/SN74H*, and -18 mA for SN54LS/SN74LS* and SN54S/SN74S*.

Not more than one output should be shorted at a time, and for SN54H/SN74H, SN54LS/SN74LS* and SN54S/SN74S*, duration of output short circuit should not exceed one second.

supply current[†]

TYPE	I _{CCH} (mA) Total with outputs high		I _{CCL} (mA) Total with outputs low		I _{CC} (mA) Average per gate (50% duty cycle)
	TYP	MAX	TYP	MAX	TYP
'08	11	21	20	33	3.88
'H11	18	30	30	48	8
'H21	12	20	20	32	8
'LS08	2.4	4.8	4.4	8.8	0.85
'LS11	1.8	3.6	3.3	6.6	0.85
'LS21	1.2	2.4	2.2	4.4	0.85
'S08	18	32	32	57	6.25
'S11	13.5	24	24	42	6.25

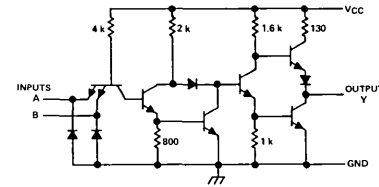
[†]Maximum values of I_{CC} are over the recommended operating ranges of V_{CC} and T_A; typical values are at V_{CC} = 5 V, T_A = 25°C.

switching characteristics at V_{CC} = 5 V, T_A = 25°C

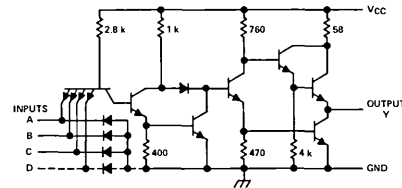
TYPE	TEST CONDITIONS#	tPLH (ns) Propagation delay time, low-to-high-level output			tPHL (ns) Propagation delay time, high-to-low-level output		
		MIN	TYP	MAX	MIN	TYP	MAX
'08	C _L = 15 pF, R _L = 400 Ω		17.5	27		12	19
'H11, 'H21	C _L = 25 pF, R _L = 280 Ω		7.6	12		8.8	12
'LS08, 'LS11 'LS21	C _L = 15 pF, R _L = 2 kΩ		8	15		10	20
'S08, 'S11	C _L = 15 pF, R _L = 280 Ω		4.5	7		5	7.5
	C _L = 50 pF, R _L = 280 Ω		6			7.5	

[#]Load circuit and voltage waveforms are shown on pages 3-10 and 3-11.

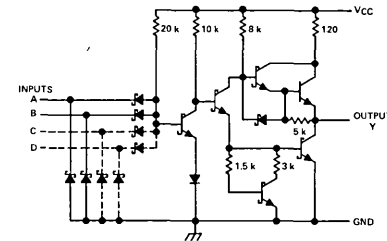
schematics (each gate)



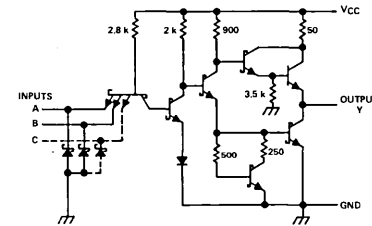
'08 CIRCUITS



'H11, 'H21 CIRCUITS



'LS08, 'LS11, 'LS21 CIRCUITS



'S08, 'S11 CIRCUITS

Resistor values shown are nominal and in ohms.